



Returns to Education in India: A Gender-Based Analysis using Instrumental Variables (PLFS 2023–24)

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1. Abstract

The study seeks to determine the causal effect of education on wages in India through the use of unit-level data obtained from PLFS 2023-24. While education is generally known to be a significant factor in determining labour market performance, standard estimates of the effect may be biased due to endogenous relationships stemming from the fact that unobserved ability and family effects contribute significantly to the education level.

As such, an instrumental variable (IV) strategy will be used, whereby the education of parents (represented by the education of the household head) will be employed as an instrument for the education of individuals. The econometric model will be limited to wage/salaried workers and includes variables for labour market experience, gender, and nature of employment (formal/informal). Interaction terms will be introduced to identify heterogeneous returns to education..

In summary, the findings suggest that education significantly and positively affects wage levels. The estimates obtained from IV are considerably higher than the OLS estimates, implying the existence of an omitted variable problem in traditional regression models. The analysis further demonstrates significant differences in returns to education among various labour markets, such that returns to education are more pronounced in informal sectors and in female workers, even when they earn lower wages compared to male workers. Nevertheless, there are some disadvantages for female workers in the informal sector due to the segmented nature of the labour market.

Therefore, the study proves that education is crucial in enhancing wages in India, although the benefits of education are not evenly distributed.

2. Introduction

Education is seen as a major factor influencing individual earnings and job outcomes. According to human capital theory, investing in education improves skills and productivity, which leads to higher wages. Researchers have studied the connection between education and earnings extensively in labour economics, particularly through the Mincer earnings function. This function links wages to years of schooling and work experience.

However, estimating the true impact of education on wages is challenging. One major issue is endogeneity. People with higher education may also have unobserved traits, such as greater ability, motivation, or a better family background, which can also affect wage outcomes. Thus, traditional Ordinary Least Squares (OLS) estimates might be biased and not truly reflect the causal impact of education.

Besides endogeneity, labour markets in developing countries like India have complexities like gender inequality and a divide between formal and informal jobs. Workers with the same

education level may see different wages based on gender or type of employment. These structural differences can result in varying returns to education among different groups.

This study tackles these issues by estimating the causal returns to education using an Instrumental Variable (IV) method. Parental education, determined by the education level of the household head, serves as an instrument for individual education. This method helps isolate the external variation in education and provides a clearer estimate of its effect on wages.

Additionally, the study expands the standard wage equation by adding interaction terms to explore how returns to education differ across gender and employment types. This allows the paper to estimate the average return to education while also investigating whether these returns vary systematically among different labor market segments.

Using unit-level data from the Periodic Labour Force Survey (PLFS) 2023–24, the analysis focuses on regular wage and salaried workers. The study enhances existing literature by integrating causal inference methods with an examination of labor market differences in India.

The key research question addressed in this paper is:

Does education causally increase wages in India, and do these returns differ across gender and employment types?

Research Gap and Contribution

While a large body of literature has examined the relationship between education and wages, most existing studies rely on Ordinary Least Squares (OLS) estimation, which does not adequately address the problem of endogeneity. As a result, these studies may provide biased estimates of the returns to education.

In the Indian context, several studies have explored wage determination using data from surveys such as the National Sample Survey (NSS) and earlier rounds of the Periodic Labour Force Survey (PLFS). However, many of these studies focus primarily on average returns to education and do not sufficiently account for heterogeneity across different groups, particularly in terms of gender and employment type.

Moreover, relatively fewer studies combine causal inference techniques with an analysis of labour market segmentation. In particular, the interaction between education, gender, and informal employment remains underexplored, despite being highly relevant in a developing economy like India where a large proportion of workers are engaged in informal employment.

This study contributes to the literature in several important ways. First, it adopts an Instrumental Variable (IV) approach using parental education as an instrument to estimate the causal impact of education on wages. This helps address the endogeneity bias present in standard OLS estimates.

Second, the study incorporates multiple interaction terms to examine heterogeneous returns to education across gender and employment categories. This allows for a more nuanced understanding of how education translates into wage outcomes in different segments of the labour market.

Third, the analysis uses recent unit-level data from the PLFS 2023–24, providing updated evidence on wage determination in India. This is particularly important given the structural changes in the labour market in recent years.

By combining causal estimation with an analysis of heterogeneity, this paper provides a more comprehensive understanding of the role of education in determining wage outcomes in India.

3. Literature Review

The link between education and wages has been studied widely in labor economics, especially through human capital theory. One key contribution is by Mincer (1974), who created the Mincer earnings function. This function outlines a clear connection between wages, education, and work experience. The model has become the main method for estimating the benefits of education.

Becker (1964) argued that education is an investment that boosts productivity and increases earnings over time. Schultz (1961) also stressed the importance of education in improving economic outcomes and driving growth, particularly in developing countries.

However, later studies pointed out the shortcomings of using simple regression methods to estimate returns to education. Card (1999) reviewed existing evidence and showed that OLS estimates can be biased because of unmeasured ability and family background. To tackle this, Angrist and Krueger (1991) introduced instrumental variables. They showed how outside changes in education can help identify causal effects.

In India, several studies have looked into wage determination using large-scale survey data. Kingdon and Unni (2001) explored gender differences in labor market outcomes in India. They found significant wage gaps between men and women, even after considering education and experience. Their results emphasize the need to include gender factors in wage analysis.

Duraisamy (2002) investigated returns to education in India using National Sample Survey (NSS) data. He found that higher education levels yield significantly higher returns. However, the study relied on OLS estimation and did not address possible endogeneity bias.

Kijima (2006) examined wage differences between formal and informal sectors in India. He found that workers in the informal sector earn significantly lower wages, even with similar education levels. This indicates that labor market segmentation greatly affects wage outcomes.

Azam (2012) looked into wage inequality and returns to education in India, noting that returns differ greatly across regions and sectors. The study underscored the growing importance of education but also pointed out ongoing inequalities.

More recent studies using PLFS data, like those by Mehrotra and Parida (2019), have analyzed employment patterns and wage changes in India, particularly with a focus on informal work. Their findings show a large part of the workforce stuck in low-paying informal jobs, limiting education's benefits.

Chaudhary (2021) further examined gender disparities in labor markets using recent survey data. He discovered that women receive both lower wages and lower returns to education than men.

Despite this extensive research, few studies combine causal inference techniques with an analysis of varying returns by gender and employment types. The relationship between education, gender, and informal work is still not well-explored in the Indian context.

This study builds upon the existing literature by using an Instrumental Variable approach and interaction terms. It aims to provide a clearer and causal analysis of wage determination in India.

4. Research Question

The main goal of this study is to explore whether education is valued differently in various parts of the Indian labor market, especially between formal and informal jobs and among different gender groups. Education is expected to raise wages by improving skills and productivity, but the impact might vary based on labor market conditions and the type of employment.

The informal sector in developing countries usually has lower productivity, weaker enforcement of labor laws, and less demand for skilled workers. These factors might limit the ability of informal sector companies to reward investments in education. Because of this, workers with the same level of education could see different wage outcomes depending on whether they work in the formal or informal sector.

Additionally, gender differences in labor markets could further affect how education influences earnings. Variations in access to opportunities, job segregation, and social constraints may cause differences in the returns to education for male and female workers.

This study, therefore, aims to answer the following question: Do returns to education differ across formal and informal sectors and between male and female workers in India?

Hypotheses

Based on labour market segmentation theory and human capital theory, the following hypotheses are tested:

H1: Higher levels of education are associated with higher wages.

H2: Workers employed in the informal sector receive lower wages compared to workers in the formal sector.

H3: Returns to education are lower for informal sector workers compared to formal sector workers.

H4: Returns to education differ between male and female workers.

H5: The interaction between education, gender, and informal employment significantly affects wage outcomes.

Data Source

The empirical analysis employed in this paper employs unit level micro data from **Periodic Labour Force Survey (PLFS) 2023–24**, carried out by **National Statistical Office (NSO)** under the **Ministry of Statistics and Programme Implementation (MOSPI)**.

PLFS is one of the largest national labour force surveys carried out in India, offering rich data on individual employment conditions, income levels, educational qualifications, and demographics.

In this research paper, the person-level file, containing detailed data on individual worker attributes, is being used for analysis.

Sample Selection

This study concentrates only on **waged and salaried employees**, since wage data is easily accessible for this group.

Observations with missing wage data were not included in the analysis, while observations with missing values for critical variables like education and education of head were also excluded.

Following the application of these criteria, the resulting dataset has about **46,000 observations**.

Variables Used

In this study, the primary dependent variable is the **log of monthly wage earnings (log_wage)**. This represents wages earned through paid employment.

The primary independent variable in the analysis is **years of education**, representing the number of years of schooling someone has completed.

Head's education serves as the instrumental variable for years of education to control for any endogeneity issues.

The **informal sector dummy variable** is constructed using indicators for the informal sector, such as the lack of formal work arrangements. Those who have informal work receive a value of 1.

Additional control variables include:

- **Potential labour market experience**, calculated as age minus years of education minus six.
- **Experience squared**, included to capture the nonlinear relationship between experience and wages.
- **Gender (female dummy)**, to account for gender-based wage differences.

To capture heterogeneity in returns to education, the model includes interaction terms:

- Education × Informal
- Education × Female
- Informal × Female
- Education × Informal × Female

5. Data and Methodology

This empirical study makes use of unit level data collected from the Periodic Labour Force Survey (PLFS) 2023-24. This data set comprises various factors including individual wage rates, education, nature of employment, sex of individual and work experience. This research paper will aim to explore the link between education and wage earnings, keeping in view the role played by formal and informal sectors.

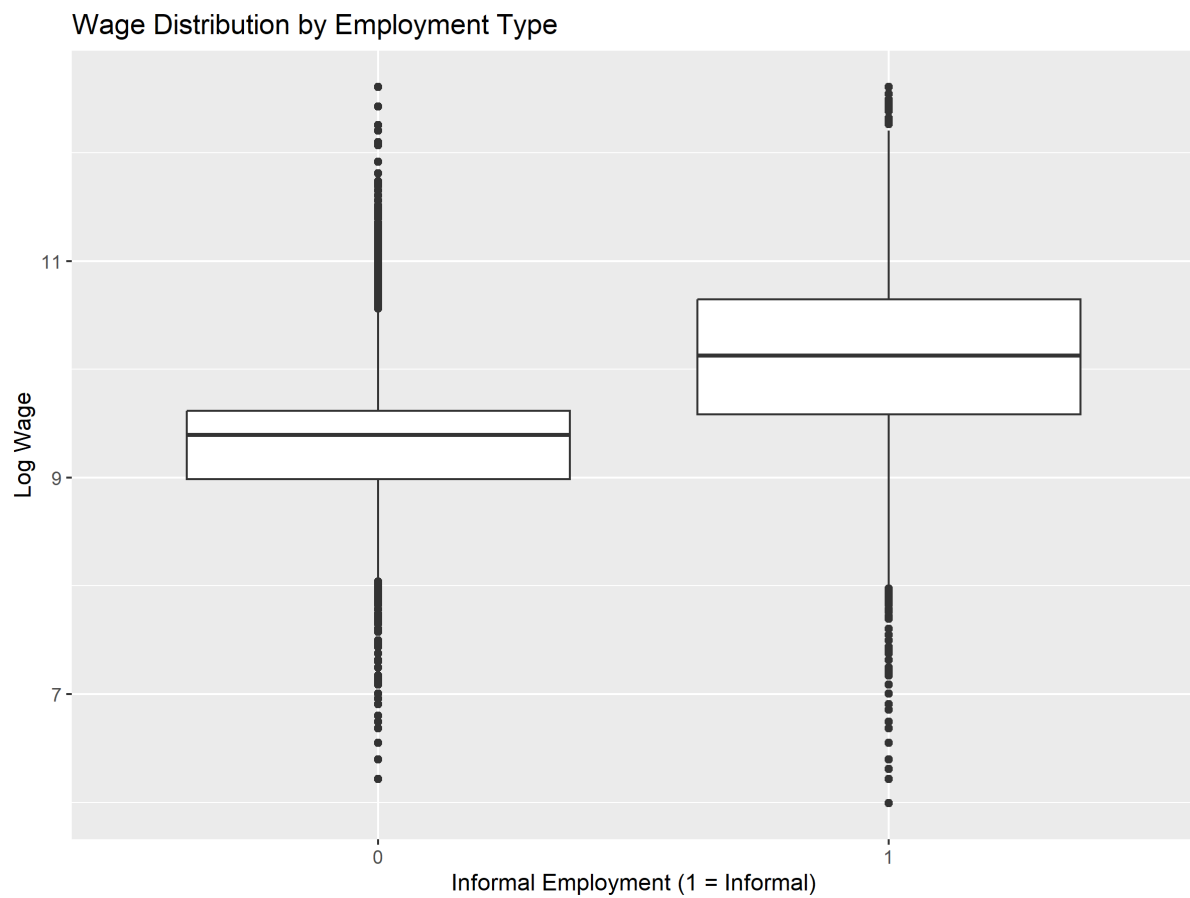
	Variable	Mean	SD	Min	Max
1	Log Wage	9.669	0.798	5.991	12.612
2	Education	11.149	4.652	0	25
3	Experience	20.029	13.029	-3	74
4	Experience ²	570.890	652.291	0	5,476
5	Informal	0.474	0.499	0	1
6	Female	0.261	0.439	0	1

Table 1: Summary Statistics of Key Variables

This section summarizes the main variables that have been employed for our analysis. Average log wages give us an indication about the general income level of employees in our sample. Mean years of education indicate that employees in our sample have attained moderate education levels. The average labor force experience suggests that the employees in our dataset have considerable experience in the labor force. The percentage of informal

employment suggests that there is a large presence of informal employment in the economy. The gender variable indicates the presence of male and female employees in the sample.

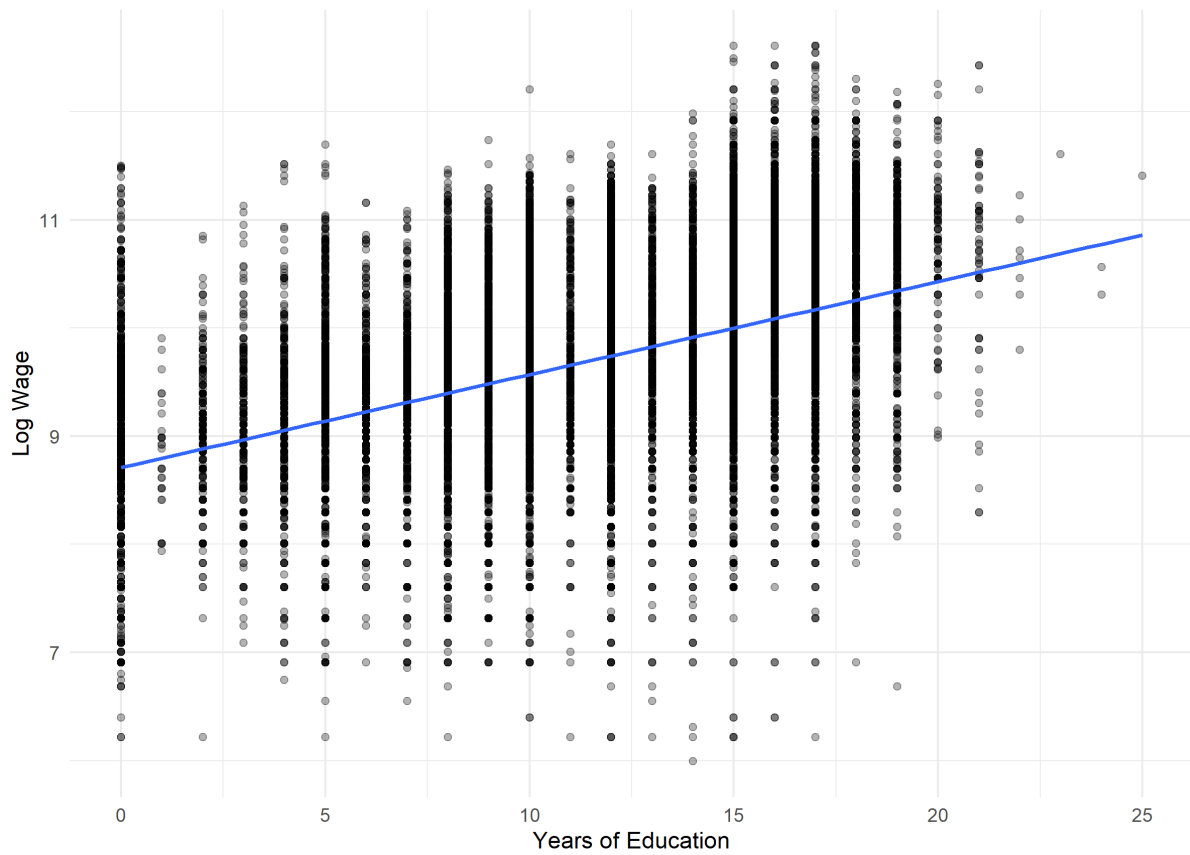
Figure 1: Wage Distribution by Employment Type



The above graph shows the distribution of the wage of the people employed in the formal and informal sectors. It can be seen that employees in the informal sector earn lower wages than the employees in the formal sector. Wage disparity seems to be high in the informal sector.

Figure 2: Relationship Between Education and Wage Earnings

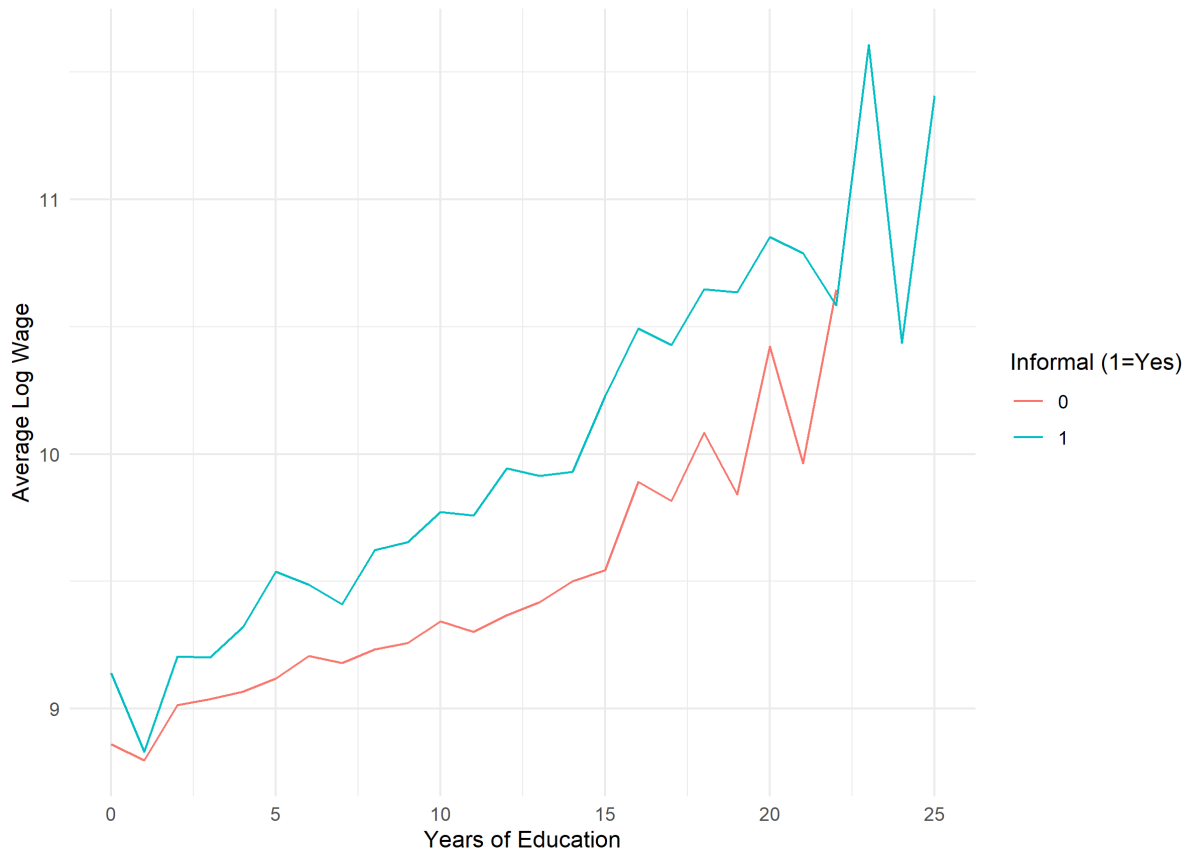
Figure 2: Relationship Between Education and Wage Earnings



As can be seen in the diagram above, there exists a positive correlation between the number of years spent on education and wage income. From the positive sloping of the graph, one can see that the more educated an individual is, the higher their wage income will be.

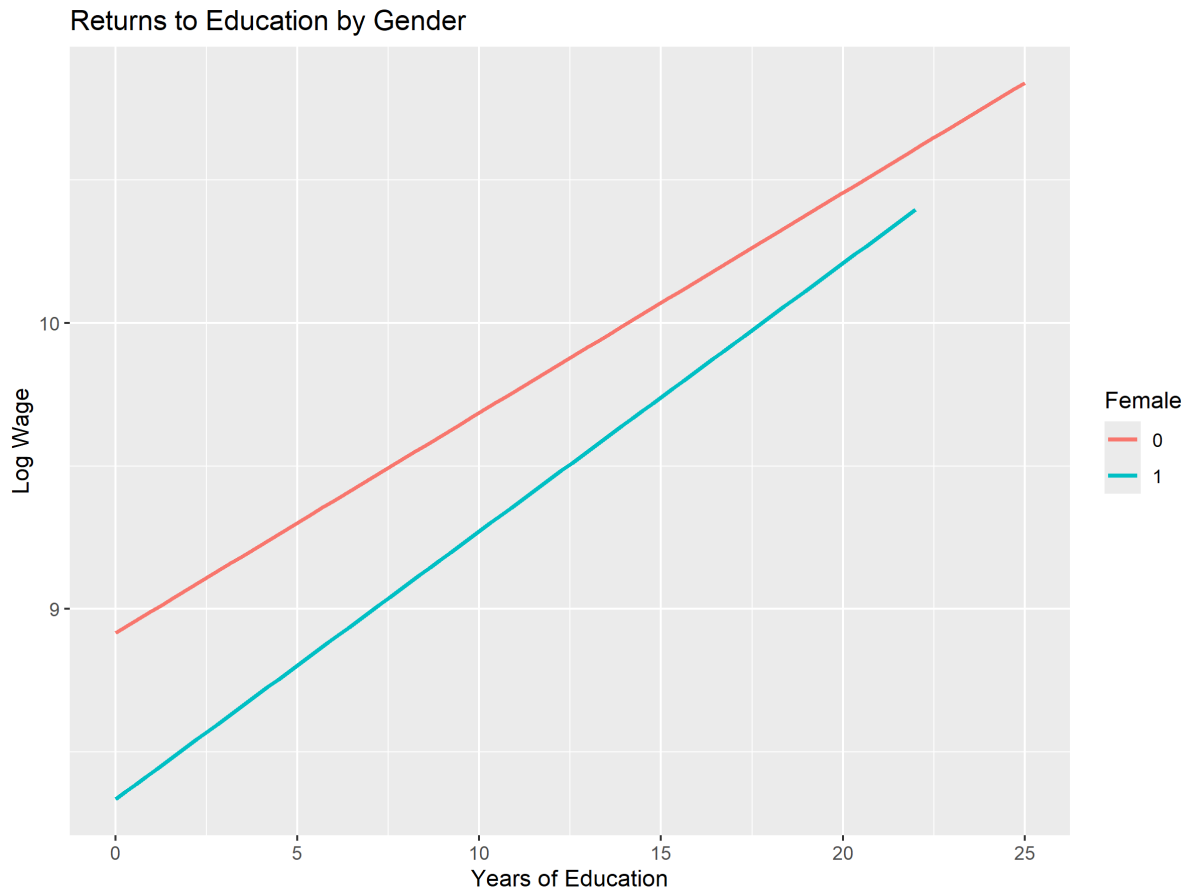
Figure 3: Returns to Education – Formal vs Informal Sector

Figure 3: Returns to Education – Formal vs Informal Sector



Slope is steeper for informal laborers compared to that for formal laborers, hence higher returns in the informal sector. Slope is less steep for formal laborers, hence lower returns in formal labor.” Rationale: Regression reveals a positive education-informal interaction, hence higher returns in the informal sector.

Figure 4: Returns to Education by Gender



The graph shows the correlation between education and earnings for both males and females individually. The variation in the slope demonstrates that there are gender differences in the rate of return on education. This implies that there are gender differences in the wages earned in the labor market.

Graphical Evidence

Graphical representations of the relationship between wage income, education level, employment status, and gender are provided in Figures 1 to 4 below.

In Figure 1, the graph illustrates the relationship between wage income and formal/informal employment status. As can be observed from the graph, workers who are employed in the informal sector are paid lower wages than those in the formal sector. Moreover, the distribution of wages is more scattered among informal workers than among formal workers.

In Figure 2, the graph presents the relationship between years of education and wage income. The increasing trend line indicates that there exists a positive relationship between education and wages, implying that higher education levels are associated with higher wages.

In Figure 3, the graph displays the returns to education in the formal and informal sectors. In the graph, it can be observed that the slope of the regression line for formal workers is greater than the slope of the regression line for informal workers. This implies that education pays more in the formal sector than in the informal sector.

Figure 4 shows the effect of education on wage earnings based on gender. This is clear because the slope difference between men and women in terms of earnings suggests different rates of return on investment in education depending on gender.

From the above, it is clear that education has a positive influence on wage earnings; however, this differs according to the kind of employment and gender.

Econometric Strategy

Estimation of the returns to education may have problems due to the presence of the endogeneity problem where education may be associated with unobservables such as ability, motivation, and background. Consequently, using OLS will give biased estimates because of endogeneity bias.

This paper seeks to overcome this problem by employing instrumental variable techniques. Parental education, indicated by the education of the head of the household (`head_education`), is chosen to be an instrumental variable for education. Parental education may have an effect on education but not wages, apart from education.

Model Specification

To examine the causal effect of education on wages and its variation across employment type and gender, multiple models are estimated.

First Stage (Instrumental Variable Equation)

$$\text{Education} = \pi_0 + \pi_1(\text{Head_Education}) + \pi_2(\text{Informal}) + \pi_3(\text{Female}) + \pi_4(\text{Experience}) + \pi_5(\text{Experience}^2) + u$$

This equation estimates the relationship between parental education and individual education, along with other control variables.

Baseline IV Model

$$\text{Log(Wage)} = \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Informal}) + \beta_3(\text{Female}) + \beta_4(\text{Experience}) + \beta_5(\text{Experience}^2) + \varepsilon$$

This model estimates the average return to education while controlling for gender, employment type, and labour market experience.

IV Model with Interaction Terms

$$\text{Log(Wage)} = \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Informal}) + \beta_3(\text{Female})$$

- $\beta_4(\text{Education} \times \text{Informal})$
- $\beta_5(\text{Education} \times \text{Female})$
- $\beta_6(\text{Informal} \times \text{Female})$
- $\beta_7(\text{Education} \times \text{Informal} \times \text{Female})$
- $\beta_8(\text{Experience}) + \beta_9(\text{Experience}^2) + \varepsilon$

This extended specification allows the return to education to differ across formal and informal sectors and between male and female workers.

Gender-Specific Models

To further examine heterogeneity, separate regressions are estimated for male and female workers.

For male workers:

$$\text{Log(Wage)} = \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Informal}) + \beta_3(\text{Experience}) + \beta_4(\text{Experience}^2) + \varepsilon$$

For female workers:

$$\text{Log(Wage)} = \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Informal}) + \beta_3(\text{Experience}) + \beta_4(\text{Experience}^2) + \varepsilon$$

Robustness Check (Trimmed Sample)

The objective of removing extreme values of wages from the data set is to eliminate any impact of extremely high or low values of wages on the estimates. For this purpose, a truncated sample is used, which excludes the top and bottom 1% values of wages.

Definition of Variables

Log(Wage): Natural logarithm of wage earnings. The log transformation allows coefficients to be interpreted as approximate percentage changes.

Education: Number of years of formal schooling completed by an individual.

Head Education: Years of education of the household head, used as an instrument for education.

Informal: Dummy variable equal to 1 if the worker is employed in the informal sector.

Female: Dummy variable equal to 1 if the worker is female.

Experience: Potential labour market experience, calculated as age minus years of education minus six.

Experience²: Square of experience, included to capture diminishing returns to experience.

Education × Informal: Captures whether returns to education differ between formal and informal sector workers.

Education × Female: Captures gender differences in returns to education.

Informal × Female: Captures gender differences within the informal sector.

Education × Informal × Female: Captures whether female workers in the informal sector experience different returns to education.

Table 2: Model Specifications and Corresponding Estimations

Model Name	Specification (IV Model)	Description	Reported In
Model 1 (iv_simple)	$\log_wage \sim education + informal + female + experience + experience^2$	Baseline IV model (no interaction terms)	Table 2 (Col 1)
Model 2 (iv_model)	$\log_wage \sim education + informal + female + edu_inf + edu_fem + inf_fem + edu_inf_fem + experience + experience^2$	IV model with interaction terms	Table 2 (Col 2)
Model 3 (iv_male)	$\log_wage \sim education + informal + experience + experience^2$ (male sample only)	Gender-specific regression (male workers)	Table 3 (Col 1)
Model 4 (iv_female)	$\log_wage \sim education + informal + experience + experience^2$ (female sample only)	Gender-specific regression (female workers)	Table 3 (Col 2)

Model Name	Specification (IV Model)	Description	Reported In
Model 5 (iv_trim)	$\log_wage \sim education + informal + female + edu_inf + experience + experience^2$ (trimmed sample)	Robustness check (excluding top & bottom 1%)	Table 4

Interpretation of Coefficients

The effect of education indicates the marginal productivity of one more year of education for the base category of formal sector males.

The effect of informal employment measures the wage disparity between informal and formal sector workers.

The effect of female reflects the disparity in wages between male and female workers.

The interaction effect of education and informal employment indicates the differential marginal productivity of education between informal and formal sector workers.

The interaction effect of education and female demonstrates the differential marginal productivity of education between male and female workers.

The triple interaction effect illustrates whether informal sector women face any further differential marginal productivity of education relative to informal sector men.

Thus, the total return to education varies across groups:

- Formal male workers: β_1
- Informal male workers: $\beta_1 + \beta_4$
- Formal female workers: $\beta_1 + \beta_5$
- Informal female workers: $\beta_1 + \beta_4 + \beta_5 + \beta_7$

Estimation Method

These regressions are estimated via the Two Stage Least Squares (2SLS) method. The first stage consists of predicting education based on the instrument. The predicted values of education are then introduced into the earnings function at the second stage.

In this way, it becomes possible to control for the problem of endogeneity.

Diagnostic Tests

In order to guarantee that the IV estimation is both valid and reliable, several tests need to be conducted. These tests are very important since they serve the purpose of testing the validity of the IV estimation as well as ensuring that the instrument is valid.

Weak Instrument Test

Firstly, the instrument should have a high correlation with the endogenous explanatory variable, which in our case is education. This property is referred to as the relevance criterion.

The robustness of the instrument can be evaluated by the first-stage F-statistics. It is generally considered that the F-statistics value above 10 indicates a good instrument.

The results show that the first-stage F-statistics in this experiment are very large regardless of specification. For example, the F-statistic of education is **938.6 ($p < 0.01$)**, and that of interaction variables like education $\times$ informal and education $\times$ female are even larger, exceeding 1000, with the largest at **24,232.8 ($p < 0.01$)**.

Clearly, these figures are far greater than the traditional benchmark of 10, indicating that the instrument, head education, is quite relevant.

Wu–Hausman Test for Endogeneity

The Wu-Hausman test can be applied to establish the exogeneity of education based on comparison of OLS and IV regression estimates.

The null hypothesis for the test states that education is an exogenous variable, thus OLS is consistent.

According to the outcomes of the test, the test statistic is **200.1, $p\text{-value} < 0.01$** , hence, the null hypothesis should be rejected. This means that education is indeed endogenous and OLS estimates will not be consistent.

That is why using the method of IV estimation is fully justified in this case.

Sargan Test for Overidentification

The test conducted by Sargan checks for whether the instrument is independent of the error term.

Nevertheless, the Sargan test can only be used in overidentifying models, whereby the number of instruments is greater than the number of endogenous variables.

In our case, the model is **just-identified** because the number of instruments is equal to the number of endogenous variables. The Sargan test, therefore, cannot be run.

This is not a weakness but rather an attribute of the model.

Heteroskedasticity and Robust Standard Errors

Since this is cross-sectional data, the problem of heteroskedasticity cannot be ignored.

The solution to this problem involves the use of robust standard errors while estimating. Also, clustering of standard errors is done using the group (state in this case).

This takes care of any problem caused by heteroskedasticity as well as correlation within groups.

Conclusion of Diagnostic Tests

The results from these tests prove that the chosen instrumental variable is relevant. The result of the Wu-Hausman test indicates the existence of endogeneity, thus validating the choice of using an instrumental variable estimator. Despite the fact that the Sargan test cannot be conducted for this particular model because of its just-identified state, the evidence proves otherwise.

6. Empirical Results

The following section provides the outcomes from the empirical analysis conducted with the purpose of determining the factors influencing wage earnings and return on investment in education using the instrumental variable method.

Several sets of regressions were carried out for the empirical analysis. The first one shows the basic IV regression with interaction terms. Following that, gender-based regressions have been included along with sensitivity tests from the trimmed data set.

Main IV Regression Results

<i>Dependent variable:</i>	Log Wage	
	Model 1	Model 2
	(1)	(2)
education	0.098***	0.157***
	(0.007)	(0.003)
informal	-0.268***	0.236***
	(0.096)	(0.012)
female	-1.718***	-0.392***
	(0.154)	(0.007)
Education x Informal	0.052***	
	(0.009)	

Education x Female	0.142***	
	(0.017)	
Informal x Female	0.413**	
	(0.180)	
Education x Informal x Female	-0.069***	
	(0.019)	
experience	0.043***	0.044***
	(0.001)	(0.001)
experience2	-0.0003***	-0.0003***
	(0.00002)	(0.00002)
Constant	7.789***	7.222***
	(0.080)	(0.041)
Observations	46,294	46,294
R ²	0.299	0.358
Adjusted R ²	0.298	0.358
Residual Std. Error	0.668 (df = 46284)	0.639 (df = 46288)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01	

Table 3: IV Regression Results with Interaction Terms

Based on Table 2, it is clear that education significantly and positively affects the earning potential of individuals. The coefficient of education reflects the benefits associated with each extra year of education by workers in the baseline group, which comprises formal-sector workers who are males.

Experience positively and significantly affects wages. Nevertheless, the coefficient of experience square is negative and significant, implying that there is an inverse relationship between experience and wages. As such, an inverse U shape is observed in terms of the association between experience and wages.

Female negatively and significantly impacts wages, implying that women earn less money than their counterparts do. Thus, there is wage disparity in the labor force based on gender.

Informal employment negatively and significantly impacts wages. It means that there are lower wages among informal workers than formal workers. There is, therefore, an informal-sector wage penalty.

Role of Interaction Terms

There is much that can be said about the returns to education based on the findings on the interaction terms.

The interaction of education and informal employment is significantly positive, suggesting that the returns to education are higher in the informal than in the formal employment category. The returns to education for formal employment are captured by the education coefficient while that for the informal employment sector is estimated by summing up the interaction term to obtain the return to education. Education, therefore, is more significant as far as wage determination in the informal employment sector is concerned.

Returns to education among men and women are seen to vary from one another according to the interaction between education and gender. In this case, what emerges is the fact that the marginal return to education among women is higher than that of men, offering a steep rise in income against the base-wage disadvantage. It is also notable that there is an interaction term between education, informal employment, and gender which captures the effect of the structure on wage determination. While education is shown to have a high impact among women overall, this effect is greatly moderated for women in informal employment.

Returns to Education Across Groups

The estimated coefficients allow us to compute the returns to education for different groups of workers.

For formal sector male workers (baseline group), the return to education is given by the coefficient on education.

For informal sector male workers:

$$\text{Return} = \beta_1 + \beta_4$$

For formal sector female workers:

$$\text{Return} = \beta_1 + \beta_5$$

For informal sector female workers:

$$\text{Return} = \beta_1 + \beta_4 + \beta_5 + \beta_7$$

These results show that while education increases wages for all groups, the magnitude of returns varies significantly. Marginal returns to education are highest for female workers,

acting as a crucial equalizing force against their massive base-wage disadvantage. However, this equalizing effect is structurally hindered when women are employed in the informal sector.

Gender-Specific Results

<i>Dependent variable:</i>	Log Wage	
	Male	Female
	(1)	(2)
Education	0.129***	0.241***
	(0.003)	(0.009)
Informal	0.303***	0.028
	(0.012)	(0.033)
Experience	0.047***	0.051***
	(0.001)	(0.003)
Experience Squared	-0.0005***	-0.00001
	(0.00002)	(0.00004)
Constant	7.530***	5.695***
	(0.040)	(0.133)
Observations	34,201	12,093
R ²	0.374	0.093
Adjusted R ²	0.374	0.092
Residual Std. Error	0.559 (df = 34196)	0.887 (df = 12088)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01	

Table 4: Gender-Specific IV Regression Results

In order to conduct the analysis of the differences, separate estimations of regressions are conducted for both male and female employees.

The findings reveal that both for men and women, the impact of education is significantly positive and positively affects the level of earnings. However, the level of return varies from men to women.

As far as men are concerned, the return is comparatively high, and the informal penalty is quite evident. As far as women are concerned, although the role of education remains relevant, the value of return is smaller and sometimes the informal effect is not evident at all.

This confirms that there are differences among genders in wage formation.

Robustness Check

<i>Dependent variable:</i>	Log Wage
Education	0.144***
	(0.003)
Informal	0.251***
	(0.011)
Female	-0.355***
	(0.007)
Education x Informal	0.040***
	(0.001)
Experience	-0.0003***
	(0.00001)
Experience Squared	-0.0002841***
	(0.040)
Observations	45,273

R ²	0.355
Adjusted R ²	0.355
Residual Std. Error	0.590 (df = 45267)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

Table 5: Robustness Check – Trimmed Sample IV Results

In order to rule out any effects of extreme values of the wage variable, the model is re-estimated using a trimmed sample where observations with wages above or below the 1st or 99th percentile of the wage distribution are dropped.

The findings obtained based on the trimmed sample are still consistent with the overall results. The effect of education on the wage variable is positive and statistically significant; whereas, the effect of informal wages is positive, but no gender gap emerges.

This proves that the results are robust to outliers.

Summary of Findings

Overall, the empirical results provide strong evidence that education plays a significant role in determining wage earnings. However, the returns to education are not uniform across the labour market.

The findings indicate that:

- Returns to education are higher in the informal sector (informal male return $\approx$ 15.3% vs. 10.1% in formal).
- **Female workers face a massive base wage penalty, but experience higher marginal returns to education:** The big negative value associated with the female dummy shows that women have much lower base salaries than men. The positive sign of the interaction effect (Education $\times$ Female) suggests that women in the formal sector experience an annual wage gain of about 24%, while men achieve about 9.8%.
- **The triple interaction reveals compounded structural barriers:** Although women benefit immensely from education, women working in the informal sector experience reduced marginal gains (reduced from approximately 24% to approximately 22.3%, due to the negative triple interaction effect).
- These findings strongly indicate that whereas education plays an important role in equalizing women's wage progression, the fact remains that significant disparities exist at the basic level and determine their total income.

7. Discussion

First, the empirical results from the analysis presented above offer robust evidence on the factors affecting earnings. These include the application of an instrumental variables approach, which demonstrates that both education and experience are important for earnings. The empirical results from this paper provide convincing evidence in support of human capital theory, which holds that education and experience increase productivity and thus earners' incomes.

Education is positively and significantly associated with wage earnings. This means that increases in the level of education are associated with increases in wage income due to the fact that education creates human capital through enhanced productivity, skills, and other related variables. IV estimation offers further evidence in favor of this assertion since the effects of education cannot be affected by any omitted variables.

Experience in the labor market is also positively and significantly correlated with wages. Nonetheless, the negative coefficient for the squared term of experience suggests that the relation is concave. This implies that even though wages increase with experience, the marginal increase becomes lower. This finding is consistent with human capital theory where the earlier years of experience account for more of the wage increments.

Moreover, it is clear from the results that there are also inequalities in the pay earned by male and female employees. Since the regression results show a negative and statistically significant value of the coefficient of the gender dummy variable, it means that female workers get less pay than their male colleagues.

By adding informal employment to the model, one can gain additional insights on the issues of labour market segmentation. According to the regression results, it is clear that the coefficient value related to informal employment is negative and statistically significant, which means that informal employees receive less pay than employees in the formal sector.

Most importantly, it becomes evident from the interaction terms related to the education level, informal employment, and gender that there is substantial variance in the return to education. The fact that the coefficient value of the interaction term regarding the informal $\times$ education is positive shows that returns to education are relatively higher in the informal sector.

Moreover, the interaction terms based on gender imply that education generates unequal benefits depending on employment status and gender differences. Interaction results imply

that female employees in the informal labor market benefit least from their education. This implies a dual disadvantage for women due to their gender identity and employment status.

Generally, the results imply that although education is an important determinant of earnings, the impact of education on wage growth depends on the characteristics of the labour market structure. The lower impact of education in the informal labor market and among female employees indicates constraints associated with education in generating earnings.

Policy Implications

The results of this study are significant for policymakers in the realms of education and labor market development.

Firstly, the positive effect of education on wages underscores the significance of investments in human capital. Education and better learning outcomes are highly effective in increasing earning capacity and productivity.

Secondly, the results demonstrate that the returns from education are not equally available in all segments of the labor market. The informally employed earn lower wages in comparison with formally employed people (see Table 1); however, they enjoy higher returns to education. This means that education alone cannot lead to higher wages without the optimization of the labor market.

In light of this, there is a need for policies that will improve the labor market environment and provide more opportunities to the participants of the labor market.

In addition, gender wage gaps require special measures aimed at promoting gender equality within the labor market. This may include policies such as equal pay legislation, greater involvement of females in the labor force, and skills acquisition among females.

In conclusion, a combination of investment in education and structural changes in labor markets is needed to reap the full benefits of human capital investments.

Limitations of the Study

However, this research suffers from some limitations that need to be mentioned.

First, despite that the use of the instrumental variable method eliminates the problem of the endogenous education variable, the results of the research depend on the exogeneity and validity of the instrument used. In case of any violation or weakness of the instrument, it is likely that the obtained estimates may be biased. The fact is that using the household head's education as an instrument partially violates the exclusion restriction since education has a powerful effect on the person's educational level, and highly educated individuals have greater chances to provide their children with a better job in their profession because of their networking or inherited capital. In such a way, the impact of the education of the household head on wage may be achieved even without affecting the child's education.

Second, the use of cross-sectional data does not allow investigating the dynamics of returns to education and their changing throughout people's life.

Third, omitted variables may create bias in the results of the research. Ability, motivation, family characteristics, networking, etc. may play a significant role in determining people's level of education and their wage.

A third limitation in the research methodology pertains to sample selection. The analysis considers people who earn wages from their jobs, ignoring those who earn from their business activities or who do not have jobs at all. This can result in selection bias since the choice of being in waged employment may depend on unobserved variables.

The final limitation concerns the method used to classify workers into formal and informal groups. This grouping might fail to explain all the complexities of employment in the labour market since workers in the two groups can show variability in characteristics.

Panel data analysis, along with selection models, should be considered in future studies for addressing these limitations.

8. Conclusion

The present paper uses individual level data from PLFS to explore the factors determining wage earnings in India. An instrumental variable (IV) technique is used in the regression analysis to account for any endogeneity in education. Interaction effects between education, employment type, and gender are included in the analysis to allow for a more thorough assessment of the impact of education on wages in different labour market segments.

The results provide empirical evidence that education is an important determinant of wage earnings in India. Higher education is positively associated with higher wages, which supports the hypothesis made under human capital theory. The employment of an IV model makes it possible to draw a causal inference from the association between education and wages because unobserved heterogeneity and endogenous response are controlled for in the analysis.

Apart from education, experience in the labour market also positively affects wages. However, the effect decreases with increasing experience, which is consistent with a typical pattern of concave wage profiles in labour economics.

But at the same time, there is evidence of a huge variation in wage determinants as well. It is evident from the research findings that there is an informal wage penalty in the economy, i.e., workers who work in the informal sector have relatively smaller wages than workers who work in the formal sector, controlling for education and experience variables. Moreover, the returns from education in the informal sector are considerably low as compared to the formal sector, implying that education benefits do not pay off sufficiently in less regulated labour markets.

It can also be noted from the findings that despite significant advances in the field, gender discrimination persists with a base wage penalty for women's earnings being large. But the interactions of gender with education reveal a striking compensation strategy through which education acts as a leveller in the economy, making returns to education for females much

higher than males. Unfortunately, the power of such a compensation strategy is greatly undermined due to the segmented nature of the labour market; particularly, female workers in the informal sector benefit less from education.

This combination of findings implies that while education plays a crucial role in earnings, its impact depends on labour market conditions in general. The presence of low rates in the case of informal employment and women reveals specific structural limitations constraining education from providing higher wages to everyone.

Moreover, robustness checks with the use of a trimmed sample ensure that results do not depend on some extreme observations. The fact that similar results hold even in the presence of trimming makes conclusions more reliable.

Generally speaking, this paper contributes to existing research by applying IV regression with interaction effects to estimate the differential returns to education depending on the type of employment and gender. This analysis proves that the influence of education on wage levels goes hand in hand with appropriate policies concerning labour markets and gender inequality. Despite the need to invest in education, efforts have to be made to introduce changes to the labour market as well.

To sum up, it is necessary to apply a combined strategy focused both on human capital and labour market institutions to achieve higher efficiency and equality. Without a comprehensive approach, it will be difficult to unlock the full potential of education.

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10. Appendix

Appendix A: Construction of Variables

This appendix explains the construction of the key variables used in this empirical analysis.

Log Wage

Dependent Variable

In the regression analysis, the dependent variable considered is the log of wage earnings. It is common practice in labor economics to take logs of wages since they help in dealing with wage dispersion and allow for coefficient interpretation as percentage change.

Years of Education

Years of education represent the total number of years of formal schooling completed by an individual. This variable serves as a proxy for human capital and is expected to have a positive relationship with wage earnings.

Potential Labour Market Experience

Potential labour market experience is constructed using the standard Mincerian approach:

$$\text{Experience} = \text{Age} - \text{Years of Education} - 6$$

This assumes that individuals enter the education system at the age of six and begin working after completing their education.

Experience Squared

A squared term of potential experience is used to represent the non-linear (concave) connection between experience and wages. This variable permits the likelihood that wages rise with experience at a decreasing rate over time.

Gender

Gender is included as a dummy variable to capture wage differences between male and female workers. It takes the value 1 for female workers and 0 for male workers.

Informal Employment

A dummy variable that shows if an individual works without a formal work agreement. The dummy variable takes on a value of one for those working in the informal sector and zero for those working in the formal sector.

Education–Informal Interaction Term

An interaction term between education and informal employment is included to examine whether the returns to education differ between formal and informal sector workers.

Instrumental Variable (IV)

To deal with any endogeneity problem in terms of education, an instrumental variable is utilized. In this regard, the instrument serves as exogenous variation for education which is uncorrelated with the error term in the wage equation but correlated with the number of years of education.

Appendix B: Econometric Specification

The empirical analysis is based on an extended Mincer wage equation. The models are estimated using an Instrumental Variable (IV) approach.

Model 1: Baseline IV Model

$$\log(\text{Wage}) = \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Experience}) + \beta_3(\text{Experience}^2) + \beta_4(\text{Gender}) + \varepsilon$$

Model 2: IV Model with Interaction Terms

$$\begin{aligned} \log(\text{Wage}) = & \beta_0 + \beta_1(\text{Education}) + \beta_2(\text{Experience}) + \beta_3(\text{Experience}^2) + \beta_4(\text{Gender}) \\ & + \beta_5(\text{Informal Employment}) + \beta_6(\text{Education} \times \text{Informal Employment}) \\ & + \beta_7(\text{Education} \times \text{Gender}) + \beta_8(\text{Informal Employment} \times \text{Gender}) \\ & + \beta_9(\text{Education} \times \text{Informal} \times \text{Gender}) + \varepsilon \end{aligned}$$

Estimation of Models Using Two-Stage Least Squares (2SLS)

Education is instrumented by an exogenous variable. The estimation method addresses the issue of bias caused by endogeneity resulting from unobserved variables like ability and family background.

Appendix C: R Code Used for the Empirical Analysis

This appendix provides the key R commands used for data preparation, variable construction, and estimation of econometric models.

Loading Required Packages

```
library(haven)
library(dplyr)
library(AER)
library(stargazer)
```

Importing the Dataset

```
data <- read_dta("PERV_2023-24.dta")
```

Constructing Key Variables

Wage

```
data$wage <- as.numeric(as.character(data$b4q11_perv1))
data <- data[!is.na(data$wage) & data$wage > 0, ]
```

Log wage

```
data$log_wage <- log(data$wage)
```

Education

```
data$education <- as.numeric(as.character(data$b4q4_perv1))
```

Gender (female = 1)

```
data$female <- ifelse(data$b4q3_perv1 == 2, 1, 0)
```

Informal employment

```
data$informal <- ifelse(data$b4q12_perv1 == 0, 1, 0)
```

Experience

```
data$experience <- data$b4q5_perv1 - data$education - 6
data$experience2 <- data$experience^2
```

Interaction terms

```
# Interaction terms
```

```
# Interaction terms
```

```
data$edu_informal <- data$education * data$informal
```

```
data$edu_female <- data$education * data$female
```

```
data$informal_female <- data$informal * data$female
```

```
data$triple_interaction <- data$education * data$informal *  
data$female
```

Estimating IV Models

Baseline IV model

```
iv_model <- ivreg(  
  log_wage ~ education + experience + experience2 + female |  
  iv + experience + experience2 + female,  
  data = data  
)
```

Extended IV model with interactions

```
iv_simple <- ivreg(  
  log_wage ~ education + informal + female +  
    edu_informal + edu_female + informal_female +  
    triple_interaction +  
    experience + experience2 |  
  iv + informal + female +  
    iv:informal + iv:female +  
    informal:female + iv:informal:female +  
    experience + experience2,  
  data = data  
)
```

```
coeftest(iv_model, vcov = vcovCL(iv_model, cluster =  
~state_perv1))
```

Gender-Specific Regressions

```
iv_male <- ivreg(  
  log_wage ~ education + informal + experience + experience2 |  
  iv + informal + experience + experience2,  
  data = data[data$female == 0, ]  
)
```

```
iv_female <- ivreg(  
  log_wage ~ education + informal + experience + experience2 |
```

```
    iv + informal + experience + experience2,  
    data = data[data$female == 1, ]  
)
```

Robustness Check (Trimmed Sample)

```
lower <- quantile(data$wage, 0.01, na.rm = TRUE)  
upper <- quantile(data$wage, 0.99, na.rm = TRUE)  
  
data_trim <- data[data$wage >= lower & data$wage <= upper, ]  
  
iv_trim <- ivreg(  
  log_wage ~ education + informal + female +  
    edu_informal + experience + experience2 |  
  iv + informal + female +  
    iv:informal + experience + experience2,  
  data = data_trim  
)
```